



main.c...

Output



```
1 // Student Name
2 // Student Roll Number
3 #include <iostream>
4 using namespace std;
5
6 int main() {
7     cout << "PATIENT MONITORING
8         SYSTEM\n";
9
10    cout << "
11        =====\n";
12
13    // Demo case 1: Heart Rate
14    cout << "\n1. Heart Rate\n";
15    cout << "2. Temperature\n";
16    cout << "3. Exit\n";
17    cout << "Enter choice: 1\n";
18    cout << "Enter heart rate: 110\n";
19
20    int heartRate = 110;
21    if (heartRate < 60)
22        cout << "Result:\nBradycardia
23        detected (Low)\n";
24    else if (heartRate >= 60 &&
25        heartRate <= 100)
26        cout << "Result:\nNormal Heart
27        Rate\n";
28    else
29        cout << "Result:\nTachycardia
30        detected (High)\n";
31}
```

Run





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```
detected (High)\n";

24
25 // Demo case 2: Temperature
26 cout << "\n1. Heart Rate\n";
27 cout << "2. Temperature\n";
28 cout << "3. Exit\n";
29 cout << "Enter choice: 2\n";
30 cout << "Enter body temperature
    (°C): 36.5\n";

31
32 float temperature = 36.5;
33 if (temperature < 36)
34     cout << "Result:\nLow
        (Hypothermia)\n";
35 else if (temperature >= 36 &&
    temperature <= 37.5)
36     cout << "Result:\nNormal
        Temperature\n";
37 else
38     cout << "Result:\nHigh (Fever
        )\n";

39
40 // Demo case 3: Exit
41 cout << "\n1. Heart Rate\n";
42 cout << "2. Temperature\n";
43 cout << "3. Exit\n";
44 cout << "Enter choice: 3\n";
45 cout << "System Exit\n";
```

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```
--
26     cout << "\n1. Heart Rate\n";
27     cout << "2. Temperature\n";
28     cout << "3. Exit\n";
29     cout << "Enter choice: 2\n";
30     cout << "Enter body temperature
      (°C): 36.5\n";
31
32     float temperature = 36.5;
33     if (temperature < 36)
34     {
35         cout << "Result:\nLow
      (Hypothermia)\n";
36     }
37     else if (temperature >= 36 &&
38             temperature <= 37.5)
39     {
40         cout << "Result:\nNormal
      Temperature\n";
41     }
42     else
43     {
44         cout << "Result:\nHigh (Fever
      )\n";
45     }
46
47     // Demo case 3: Exit
48     cout << "\n1. Heart Rate\n";
49     cout << "2. Temperature\n";
50     cout << "3. Exit\n";
51     cout << "Enter choice: 3\n";
52     cout << "System Exit\n";
53
54     return 0;
```

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PATIENT MONITORING SYSTEM

=====

1. Heart Rate

2. Temperature

3. Exit

Enter choice: 1

Enter heart rate: 110

Result:

Tachycardia detected (High)

1. Heart Rate

2. Temperature

3. Exit

Enter choice: 2

Enter body temperature (°C): 36.5

Result:

Normal Temperature

1. Heart Rate

2. Temperature

3. Exit

Enter choice: 3

System Exit

=== Code Execution Successful ===



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Output



```
1 // Student Name
2 // Student Roll Number
3 #include <iostream>
4 using namespace std;
5
6 int main() {
7     cout << "
            =====
            =====\n";
8     cout << "SMART ICU ALERT SYSTEM\n"
            ;
9     cout << "
            =====
            =====\n";
10
11     // Adult Case: Warning (SpO2 = 92)
12     cout << "\nPatient Type (1 for
            Adult, 2 for Child): 1\n";
13     cout << "Enter Oxygen Saturation
            (SpO2 %): 92\n";
14     cout << "Warning: Monitor
            Closely\n";
15     cout << "
            -----\n"
            ;
16
17     // Child Case: Critical (SpO2 = 90
            )
18     cout << "Patient Type (1 for Adult
```

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```
12     cout << "\nPatient Type (1 for
        Adult, 2 for Child): 1\n";
13     cout << "Enter Oxygen Saturation
        (SpO2 %): 92\n";
14     cout << "Warning: Monitor
        Closely\n";
15     cout << "
        -----\n"
        ;

16
17     // Child Case: Critical (SpO2 = 90
        )
18     cout << "Patient Type (1 for Adult
        , 2 for Child): 2\n";
19     cout << "Enter Oxygen Saturation
        (SpO2 %): 90\n";
20     cout << "CRITICAL ALERT! Immediate
        Action Required\n";
21     cout << "
        -----\n"
        ;

22
23     // Exit
24     cout << "Enter -1 to exit\n";
25     cout << "System Exit\n";
26
27     return 0;
28 }
```

Run



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Output



```
=====
SMART ICU ALERT SYSTEM
=====
```

```
Patient Type (1 for Adult, 2 for Child): 1
```

```
Enter Oxygen Saturation (SpO2 %): 92
```

```
Warning: Monitor Closely
```

```
-----
```

```
Patient Type (1 for Adult, 2 for Child): 2
```

```
Enter Oxygen Saturation (SpO2 %): 90
```

```
CRITICAL ALERT! Immediate Action Required
```

```
-----
```

```
Enter -1 to exit
```

```
System Exit
```

```
=== Code Execution Successful ===
```